Element G

Our Prototype was designed to keep in mind ergonomics and functionality in one. In doing so, we have curated a prototype with a wider range of adjustability while trying to eliminate unnecessary elements and better the structural stability. The best way we found to stabilize the prototype was to minimize moving parts. First, we needed to figure out a way that we could mount the prototype to the cart while keeping in mind that our prototype had to be universal, so we opted for a leverage instead of an attachment to the cart. This leverage cannot be stress tested to find the maximum breaking point due to the unknown specificities of the wood used in the prototype. In order to make up for the lack of testing, we created precautions and padding or a lot of the contact areas of the prototype to the cart.

This leverage acts between the two wooden poles that situate through the top and bottom of the cart and the cart itself, making its forward and backward movements stellar, but lacking on the side to side and turns. In order to catch the slack of leverage, we created connectors that go in between the wooden poles.

These connectors are made out of PVC pipe, and act as both an anchor and a handle for the user. These PVC pipes are two separate systems, so we will refer to them as the anchor, the PVC situated at the bottom, and the handle, which is situated at the top.

The handle is made from four pieces of PVC pipe, four 90 degree elbow connectors, two pieces of rubber and hot glue. The 4 pieces of PVC pipe are connected using the elbows to create a rectangle with the dimensions of 16 inches x 6 inches. These elbows are glued onto certain parts of the pipe, allowing for dismemberment, but also steadiness in pushing and turning. The handle, when in its separate parts, can slide into two holes that are drilled through the wooden poles, making the handle 100% attached to the prototype. The rubber pieces are connected to the part of the handle that slides through said holes. They act as a friction anchor, to hold the PVC pipe in place and not letting the handle drop to a resting position. The handle was made with ergonomics in mind, while also being able to skillfully perform the tasks of turning, pushing, and pulling.

The anchor is a single piece of pvc that measures ~40 inches. The point of the anchor is to hold the bottom of the wooden poles to the cart using friction and leverage. The anchor goes through two holes drilled into the bottom of the wooden poles. The ends of the anchor sit on top of the bottom tray lip and stop the wooden poles from sliding down into the bottom tray and towards the user, stopping on the legs of the cart.

1) Using two wooden poles, labeled WP.A and WP.B, slide them through the pre-existing cart handle with the bottoms inside the lip of the bottom tray. The poles will be labeled on the top, and they should be situated the right way if you are able to read WP.A on the left, and WP.B on the right.

2) The handle will slide into the holes drilled into the WP's and connect the second part of the handle by sliding the elbows into the posts that they connect to, labels with red marker.

3) Once you situate the handle onto the wooden poles, raise the bottom of the prototype in order to put the anchor into the holes on the bottom of the WP's. Once the anchor is in, make sure that the anchor is set on top of the bottom tray lip and into the legs

of the cart. When pushing and pulling, the top of the handle offers the most comfortable grip due to the angle your arms can rest at. For turning, both the top and the bottom of the handle works, but the bottom of the handle works better due to it being closer to the leverage point.